

CYBER SECURITY COURSE

Penetration Testing Scoping and Reconnaissance

Professional Case Report 09

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Report date	24. septembar 2026
Case ID	CSC-2026-09
Discipline	Penetration testing
Overall risk	Medium
Evidence status	Laboratory methodology documented

Purpose

Collect attack-surface information while keeping the activity within an authorized scope and producing actionable evidence.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A pre-engagement reconnaissance exercise focused on public information, DNS data, exposed documents and code repositories. Reconnaissance produced a structured view of the attack surface without active exploitation. The most valuable defensive action is reducing unnecessary public detail and continuously monitoring exposed assets.

Case Profile

Scenario	A pre-engagement reconnaissance exercise focused on public information, DNS data, exposed documents and code repositories.
Environment	Kali Linux with public search services, DNS tools and document metadata analysis.
Tools	Kali Linux, WHOIS, DNS reconnaissance, Search engines, GitHub search, Document metadata tools

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Defined the target domain and kept a written scope.
2. Collected registration, DNS and subdomain information.
3. Reviewed public documents for paths, usernames, software names and URLs.
4. Searched public repositories for exposed configuration and secrets.
5. Recorded only information relevant to the assessment.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Public metadata exposure	Medium	Documents may reveal internal paths, usernames and technology choices.
F02	DNS attack surface	Medium	Subdomains can expose forgotten services and testing systems.
F03	Repository leakage	High	Public code history may contain credentials or sensitive configuration even after deletion.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
nmap --script dns-brute --script-args dns-brute.domain=wb3c.org  
Search public repositories for organization and domain references
```

Risk Assessment

Risk	Level	Operational impact
Information disclosure	Medium	Public data reduces attacker discovery cost
Credential exposure	High	Leaked secrets can enable direct access

Recommendations

1. Remove metadata from externally published documents.
2. Inventory and monitor all subdomains.
3. Enable secret scanning and protected branches in source repositories.
4. Review public exposure during every major release.

Knowledge Demonstrated

- OSINT
- DNS enumeration
- Metadata analysis
- Scope control

Conclusion

Reconnaissance produced a structured view of the attack surface without active exploitation. The most valuable defensive action is reducing unnecessary public detail and continuously monitoring exposed assets.

Hands On Lab Practical 09

Completed as part of the authorized cyber security course practical work.